

# An easy hand-eye calibration method for laser profile scanners in high precision applications using optimized view poses

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**Abstract**—Accurately solving the problem of hand-eye calibration is a fundamental requirement for high-precision robotic applications involving a vision sensor. Particularly in the case of 2D laser profile scanners, its complexity increases since individual 2D measurements are not sufficient to calculate the positional and rotational relationship between tool and scanner. In this work, we propose a novel method to solve the hand-eye calibration problem for a robot-guided 2D laser sensor. Our approach is easy to implement, does not require expert knowledge, uses everyday objects instead of expensive calibration pieces, and runs automatically with minimal user interaction. Furthermore, we present an algorithm to automatically determine an optimized set of view poses. We evaluate our approach on a welding robot application and compare the results to existing methods. Our method achieves an accuracy of 0.3 mm in terms of a mean absolute error metric in the robot cell shown in Fig. 1.

## I. INTRODUCTION

Industrial robots used for in-line metrology, handling tasks, or manufacturing processes like welding usually require high precision and additional sensors to adapt to their environment [1]. Extrinsic calibration, usually known as the hand-eye calibration problem, determines the position and rotation from the robot's tool to the external vision sensor. The problem is represented by mathematical equations of the form  $\mathbf{AX} = \mathbf{XB}$ , where  $\mathbf{A}$ ,  $\mathbf{X}$  and  $\mathbf{B}$  represent homogeneous transformation matrices [2]. While an intrinsic calibration improves the kinematic model of the robot and increases its positioning accuracy [3], it is not covered by this work, as the absolute accuracy is not relevant in this scope.

The calibration process for cameras that take full 2D or 3D images is relatively easy and can be considered solved [4]. However, calibrating a laser profile sensor remains challenging as these sensors only provide 2D data from static view poses [4]. As long as the orientation remains relatively unchanged and one processes only individual scans, manually measuring the transformation or applying simple optimization algorithms is sufficient. Otherwise, the robot equipped with the sensor (in case of a robot-guided sensor) or the calibration workpiece (in case of a static mounting) must be moved during the data acquisition procedure to generate 3D measurements.

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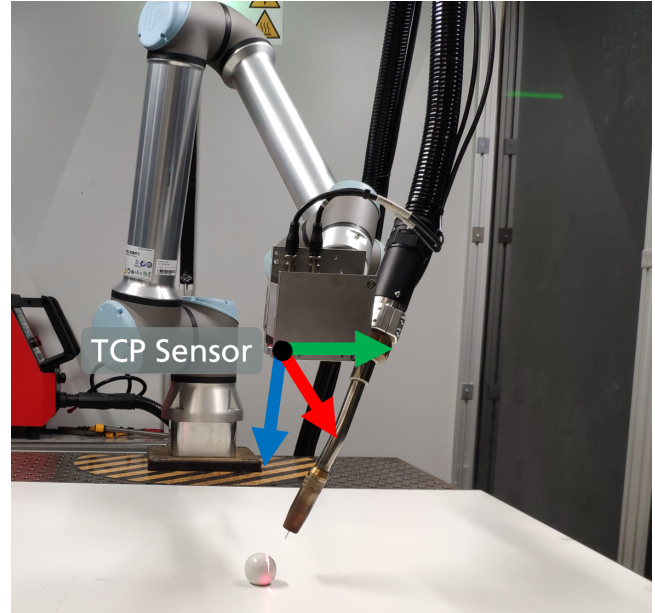


Fig. 1: The experimental setup consisting of a Universal Robot UR10e equipped with a welding gun and a 2D laser profile scanner. The transformation between flange and sensor frame (TCP Sensor) is calculated.

Such a calibration procedure must fulfill a versatile, partly contradictory set of requirements to work in a production setup: Environmental influences on the robot setup like collisions, modifications in the mounting, or simple temperature changes and wear lead to the need for re-calibration of the system. Therefore, the calibration process must run fast and without tedious manual refinement, setup time, and expert knowledge.

In particular, determining the measuring trajectories of the robot is a time-consuming process as the user must ensure that the calibration object remains in the sensor's field of view. Contrarily, the sequence program has to offer enough variance in terms of rotation and translation to allow a highly precise outcome of the calibration algorithm [4]. Simply adding more view poses increases the calibration cycle time and production downtime. Moreover, the method should avoid expensive and complicated, custom-built calibration objects.

Our approach runs automatically with minimal user input, can adapt to minor changes in the setup, automates the trajectory generation, and works with simple, everyday objects like a flat board and a table tennis ball. Moreover, we present a novel method for optimizing the view pose selection. To

the best of our knowledge, there is no existing approach that satisfies the previously elaborated requirements. The experimental scenario shown in Fig. 1 consists of a robot-guided 2D laser sensor specifically designed for welding applications but can be adapted to a statically mounted sensor and other scenarios as well.

The paper is structured as follows: Section II provides an extensive overview of related work in the field of hand-eye calibration of 2D laser sensors and optimal pose selection. Afterwards, Section III introduces a novel hand-eye calibration algorithm specifically designed for robot-guided laser profile sensors. Next, Section IV presents a meta-heuristic approach to optimize view poses for extrinsic calibration. Section V compares the procedure to existing methods. The results are elaborately discussed in Section VI and finally concluded by Section VII.

## II. RELATED WORK

Hand-eye calibration is a well-known problem in robotics and has been studied extensively in the past [5]–[7]. Following [2], the approaches are divided into three different categories: Separable closed-form solutions split the calculation of the rotational and translational part. Simultaneous closed-form solutions solve both components at the same time. Thirdly, iterative solutions apply optimization techniques instead of analytical solutions. In particular, high-precision, path-based processes like welding, painting, or industrial metrology deploy 2D laser sensors due to their high resolution, robustness against environmental influences and noise, relatively small size, and low price [1]. As described in Section I, the calibration of such sensors remains challenging. Solutions in this domain usually belong to the third category since they require complex measuring, fitting, and optimization in comparison to other sensors.

An additional camera for extending the depth profile with image data is presented by [8], with the downside of increased complexity and cost of the system. Instead of laser profile sensors, [9] propose to switch to stereo cameras which facilitate the calibration process but cannot be applied online during a highly disruptive process like welding. The so-called *SCALAR* method proposed by [10] presents a calibration method for both intrinsic kinematics and extrinsic parameters. Nevertheless, it requires a complex setup and exhibits a relatively poor accuracy in comparison to high precision applications like welding. The work described by [11] applies an Iterative Closest Point (ICP) optimization without requiring any specific calibration object but focuses on bigger workspaces with cube side lengths up to 5 m. The method is neither optimized for high accuracy in the sub-millimeter range nor achieves accuracies in this range.

The method presented by [12] introduces a calibration procedure based on a spherical object. The iterative solution approach calculates the rotational part by computing the sphere centers from linear movements above the ball, followed by the translational component based on the previously calculated rotation and a least-square optimization technique. Similarly, the approach of [13] uses a plane board as a

TABLE I: Overview of hand-eye calibration methods for 2D laser profile scanners.

| Method          | Solution            | Calibration Object       | Pose Optimization |
|-----------------|---------------------|--------------------------|-------------------|
| Lembone [10]    | intrinsic+extrinsic | single-plane             | no                |
| Peters [11]     | ICP-based           | none                     | no                |
| Liska [12]      | two-step, iterative | sphere                   | no                |
| Carlson [13]    | two-step, iterative | three planes             | no                |
| Chen [14]       | two-step, iterative | disk                     | no                |
| Santolaria [15] | intrinsic+extrinsic | multiple stepped surface | no                |
| Sharifzadeh [4] | two-step, iterative | single plane             | yes               |
| Ours            | two-step, iterative | single plane and sphere  | yes               |

calibration object. The 3D measurements of multiple, non-parallel planes serve as the basis for an iterative procedure, which applies a singular value decomposition to the covariance matrix of various points per plane and subsequent least-square optimization.

Although these methods require only simple objects and enable an easy and fast calibration procedure, both suffer from the contradictory suitability of their calibration piece. While the plane board is suited for providing multiple poses for the rotational part, the spherical point cloud offers a precise calculation of center points for the translational component. We will combine both objects in our approach. Moreover, none of these methods deal with the crucial problem of selecting good measurement poses. Programming robot view poses for hand-eye calibration manually is time-consuming and requires expert knowledge in case of 2D laser profile scanners.

Other works deploy more complex artifacts, e.g. a disk [14], an object with multiple stepped surfaces [15], or the tip of a pin [16]. Following the work of [4], highly complex objects facilitate view pose selection and increase data variance but come at the cost of simplicity and expensively manufactured objects. Their approach introduces a method that solely relies on a single-plane setup instead of multiple planes and applies a two-step, iterative algorithm.

The present work implements an automatic programming procedure based on the Robot Operating System (ROS) and a strategy to optimize measurement motions. This approach allows considering both artifact simplicity and variance in measurement data. In addition, the previously mentioned algorithms usually apply complex fitting and optimization algorithms, which makes it difficult for practitioners to use these approaches. Table I gives an overview of the presented methods for hand-eye calibration.

The work of [4] proposes a procedure to select view poses for calibration based on a circular pattern of regularly chosen sensor rotations. However, it remains unclear how the method should work robustly in a real-world scenario without knowledge of an accurate initial guess for the tool-to-sensor transformation.

In general, optimal view pose selection is a well-studied topic in different fields of applications [17]–[19]. A lot of approaches exist that optimize robot pose selection for intrinsic calibration. Those works usually apply a numerical optimization based on a Jacobian matrix [20]–[22]. The

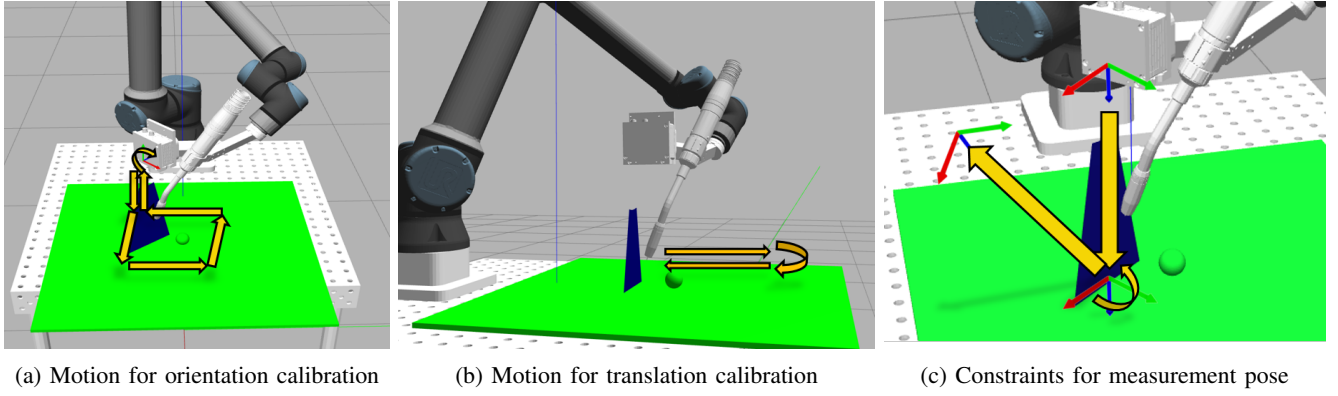


Fig. 2: Measurement motions and constraints for data acquisition and pose optimization for the two-step calibration method

singular values of a Jacobian serve as input for a so-called observability index, which determines the variance and information gain of a chosen set of robot configurations [22]–[26]. The performance of the particular observability index is difficult to compare since it heavily depends on the given robotic kinematic, its degrees of freedom, and the number of measurement poses. Section V investigates the performance of all given observability indices for our view pose optimization approach presented in Section IV.

For optimizing the chosen set of view poses, [21] draw from a pool of poses measured beforehand. This requires a large number of measurements contradicting the cycle time optimization. On the other hand, [22] first determine poses and subsequently approach them. This might lead to unsuccessful measurements due to the unknown relationship between tool and sensor.

Similar to existing approaches [20]–[22], we use non-linear optimization to choose optimal poses in a given search space. [27] propose the *DETMAX* algorithm that maximizes the determinant of the Jacobian matrix, while [20] proposes an Iterative one-by-one search (IOOPS) which is a generalized version of the *DETMAX* algorithm. To avoid being stuck in local optima, [22] extend the IOOPS algorithm with a Tabu search algorithm, which also optimizes the performance by sampling without replacement.

Contrarily to these meta-heuristic methods, [28] propose a genetic algorithm that extends such evolutionary, cross-over operations to the cost of efficiency and increased computation time. The present work applies a variant of an IOOPS algorithm, including a Tabu search algorithm designed for extrinsic calibration. To reduce the risk of choosing blind poses, we use a virtual clone of the robot cell in simulation before moving to the real-world setup.

### III. A NOVEL HAND-EYE CALIBRATION ALGORITHM FOR 2D SENSORS

Section III-A briefly introduces some terms and conventions required later on, as well as a quick overview of the calibration procedure. Subsequently, the two parts of our algorithm are presented in Section III-B and Section III-C, for orientation and translation calibration respectively.

#### A. Preliminaries

This approach considers a laser profile sensor mounted on a common six-axis robot. The equation  $\mathbf{AX} = \mathbf{XB}$  represents the hand-eye calibration problem as described in [29]. A matte white plane board with a size of 70x70 cm is placed in front of the robot. A sphere with a radius of about 20 mm is placed approximately in the center of the plane. In the following algorithm, the world frame replaces the robot base frame. Since both transformations remain constant, they can be treated equivalently.

We outline an iterative approach consisting of two optimization stages for rotation and translation, given a rough initial guess of the homogeneous transformation  $\mathbf{X}$ . For data acquisition, we assume an initial placement of the sensor such that it is roughly perpendicularly above the sphere and within the sensor's field of view. Fig. 2 pictures the motion baseline and the constraints used for automatic pose optimization. In both steps, measurements are taken from multiple viewing angles, while the viewing angle during one individual measurement, i.e. scanning a plane or sphere once, is constant. Sufficient variance in all six degrees of freedom in the calibration data set is essential.

#### B. Calibrating the orientational component

In the first part, the robot moves along a rectangle above the plane from  $n$  different sensor orientations and takes  $m$  measurements per orientation as shown in Fig. 2a. All scans  $\{s_{i,j}; i = 1 \dots n, j = 1 \dots m\}$  from each angle of view contribute to a plane fitting procedure. As presented in Fig. 3, the algorithm aims at aligning all resulting planes to the same plane and will therefore converge to an accurate solution.

The data acquisition procedure records each scan  $s_{i,j}$  with its corresponding, estimated homogeneous transformation from flange to world  $\mathbf{T}_F^W$  and groups the scans according to the current viewing angle. The scans  $s_{i,j}$  consist of 2D data points  $(x_S, z_S)$  in the sensor TCP frame.

For evaluation, a random sample consensus (RANSAC) algorithm downsamples and smooths all scans  $s_{i,j}$ . On the one hand, filtering increases accuracy since it removes sensor noise. On the other hand, it leads to higher fitting performance since it reduces the number of data points



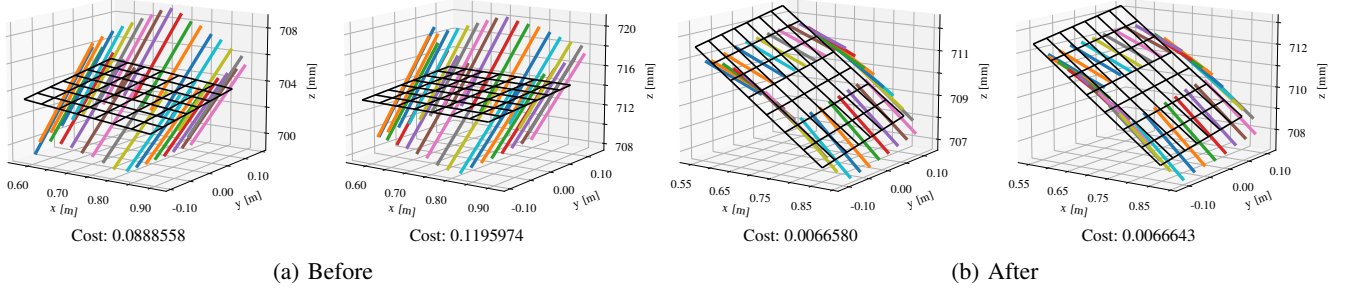


Fig. 3: Iterations of plane fitting before and after calibration for laser scans from two different view angles  $i$  displayed in world frame with their associated cost  $F_i$ . All  $j$  individual scans  $s_{i,j}$  are shown in distinct colors, while the fitted plane is shown in black. Scans of the same view angle should align to their fitted plane with minimal error.

per scan from 2000 to 50 or less. Using the homogeneous transformation from sensor to flange  $\mathbf{T}_S^F$ , the data points are transformed in each optimization step into the world frame using

$$\mathbf{p}_W = \mathbf{T}_F^W \cdot \mathbf{T}_S^F \cdot \mathbf{p}_S. \quad (1)$$

Grouping all data points of the same viewing angle results in  $p$  data points per orientation with  $\mathbf{p}_W$  (world frame) and  $\mathbf{p}_S$  (sensor frame) defined as

$$\mathbf{p}_W = \begin{bmatrix} x_W \\ y_W \\ z_W \\ 1 \end{bmatrix}, \mathbf{p}_S = \begin{bmatrix} x_S \\ 0 \\ z_S \\ 1 \end{bmatrix}. \quad (2)$$

In general, a point on a plane in 3D space with the coordinates  $(x, y, z)$  can be represented using the coordinate form

$$ax + by + cz = d, \quad a, b, c, d \in \mathbb{R}. \quad (3)$$

Next, we construct a linear equation system

$$\mathbf{A} \cdot \vec{\mathbf{x}} = \vec{\mathbf{b}} \quad (4)$$

by restructuring all data points per viewing angle  $1 \dots p$  into

$$\mathbf{A} = \begin{bmatrix} x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots \\ x_p & y_p & 1 \end{bmatrix}, \vec{\mathbf{x}} = \begin{bmatrix} a \\ b \\ d \end{bmatrix}, \vec{\mathbf{b}} = \begin{bmatrix} z_1 \\ \vdots \\ z_p \end{bmatrix}. \quad (5)$$

As the system is over-determined, we use the left pseudo inverse for optimization resulting in

$$\vec{\mathbf{x}} = (\mathbf{A}^T \cdot \mathbf{A})^{-1} \cdot \mathbf{A}^T \cdot \vec{\mathbf{b}}. \quad (6)$$

The remaining error after optimization  $\mathbf{S}$  for each data point can be calculated with

$$\mathbf{S} = \vec{\mathbf{b}} - \mathbf{A} \cdot \vec{\mathbf{x}}. \quad (7)$$

Next, we use the L2 norm to compose the cost  $F_i$  for each viewing angle using

$$F_i = \sqrt{\sum_{j=1}^p S_j^2}. \quad (8)$$

The total cost  $C$  is the sum of the L2 norm for all viewing angles and is calculated with

$$C = \sum_{i=1}^n F_i. \quad (9)$$

A trust-region reflective algorithm [30] minimizes the cost  $C$  given all considered orientations as input. The optimization uses initial values from a corresponding CAD model and is independent of the translational part of the transformation. Additionally, we introduce a boundary for optimization of  $\pm \frac{1}{4}\pi$  based on the initial values to speed up the procedure.

### C. Calibrating the translational component

In the translational calibration, the robot moves above the sphere featuring a known radius  $r$  using  $s$  different viewing angles and creates a combined point cloud from all relevant scans during the scanning motion. The motion is depicted in Fig. 2b. An accurate orientation calibration is essential to improve the accuracy of this stitching process and should therefore be done prior to this step. Next, we extract the sphere center that serves as input for the optimization of the translational part together with the corresponding transformation. Because the sphere remains fixed during calibration, the algorithm will converge to an accurate solution.

Fig. 4 illustrates the individual steps of our approach. We transform all laser scans into the frame of the first scan and then stitch them to a combined 3D point cloud. To only use relevant scan data during fitting, we remove the underlying plane by using a plane segmentation algorithm from [31] as well as other state-of-the-art point cloud filtering algorithms like radius and statistical outlier removal.

Following the work of [32], the  $l$  remaining data points are used to fit a sphere into the measured data and estimate its radius  $r_k$  and center  $\mathbf{P}_k = (x_k, y_k, z_k)$  with  $k = 1 \dots s$ , as each viewing angle results in one fit.

A single point on a sphere in a 3D space with the coordinates  $(x, y, z)$  can be represented similarly to a plane in Equation (3) with

$$(x - x_k)^2 + (y - y_k)^2 + (z - z_k)^2 = r_k^2. \quad (10)$$

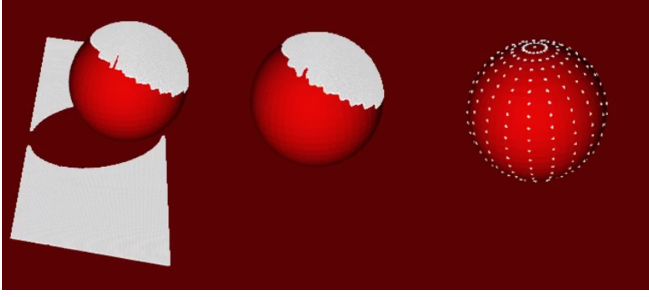


Fig. 4: Workflow of estimating data for the translational component. Scans of the sphere from one viewing angle are stitched and afterwards filtered. A least-squares fit estimates the position and size of the sphere.

Rearranging the equation to

$$\underbrace{2xx_k + 2yy_k + 2zz_k + r^2 - x_k^2 - y_k^2 - z_k^2}_{\vec{A} \vec{x}} = \underbrace{x^2 + y^2 + z^2}_{\vec{b}} \quad (11)$$

allows to convert it into a linear equation system  $\vec{A} \vec{x} = \vec{b}$  similar to Equation (5). Using the previously filtered data points  $1 \dots l$  of one viewing angle this results in

$$\begin{aligned} \vec{A} &= \begin{bmatrix} 2x_1 & 2y_1 & 2z_1 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ 2x_l & 2y_l & 2z_l & 1 \end{bmatrix}, \\ \vec{x} &= \begin{bmatrix} x_k \\ y_k \\ z_k \\ r^2 - x_k^2 - y_k^2 - z_k^2 \end{bmatrix}, \\ \vec{b} &= \begin{bmatrix} x_1^2 + y_1^2 + z_1^2 \\ \vdots \\ x_l^2 + y_l^2 + z_l^2 \end{bmatrix}. \end{aligned} \quad (12)$$

Similar to the previous subsection, the transformation from flange to world  $\mathbf{T}_F^W$  comes along with the fitting results. During calibration, another filtering removes invalid fits occurring because of noise in the sensor data using the radius  $r$ . The remaining data points are then transformed to the world frame using Equation (1). The cost function consists of the sum of the squared Euclidean distance between all center point pairs  $\mathbf{P}_k$  and is calculated with

$$C = \sum_{k=1}^{s-1} \sum_{l=k+1}^s \|\mathbf{P}_k - \mathbf{P}_l\|^2. \quad (13)$$

Analogous to Section III-B, a trust-region reflective algorithm acts as least-square method. Since the optimization problem is linear, the algorithm converges quickly compared to the orientational component and does not need any initial guess as input.

Finally, the resulting point-to-point distances determine the accuracy of the calibration. Again, since the spheres remain

fixed during calibration, these distances should be as low as possible.

#### IV. OPTIMIZING POSE SELECTION FOR EXTRINSIC CALIBRATION

Section IV-A formulates preliminary requirements before introducing a method to optimize view poses for extrinsic calibration in Section IV-B.

##### A. Preliminaries

Similar to previous works in the field of intrinsic calibration [20]–[22], the view pose optimization requires a matrix consisting of first-order partial derivatives. Instead of building a Jacobi matrix using the robot configuration and its kinematic parameters, we define a custom Jacobian for rotational and translational optimization with

$$\mathbf{J}_{\text{rot}} = \begin{bmatrix} \frac{f_{j1}}{r_x} & \frac{f_{j1}}{r_y} & \frac{f_{j1}}{r_z} \\ \frac{f_{j2}}{r_x} & \frac{f_{j2}}{r_y} & \frac{f_{j2}}{r_z} \\ \vdots & \vdots & \vdots \\ \frac{f_{jM}}{r_x} & \frac{f_{jM}}{r_y} & \frac{f_{jM}}{r_z} \end{bmatrix} \quad \mathbf{J}_{\text{trans}} = \begin{bmatrix} \frac{f_{i1}}{t_x} & \frac{f_{i1}}{t_y} & \frac{f_{i1}}{t_z} \\ \frac{f_{i2}}{t_x} & \frac{f_{i2}}{t_y} & \frac{f_{i2}}{t_z} \\ \vdots & \vdots & \vdots \\ \frac{f_{iN}}{t_x} & \frac{f_{iN}}{t_y} & \frac{f_{iN}}{t_z} \end{bmatrix} \quad (14)$$

where  $i_1, \dots, i_N$  determine the measurements of the plane,  $j_1, \dots, j_M$  refer to the spherical data,  $t_x, t_y, t_z$  and  $r_x, r_y, r_z$  are the translational and rotational part of the transformation along each axis, respectively. The algorithm uses the following different observability indices from literature:

$$\begin{aligned} O_1 &= \frac{\sqrt[4]{\sigma_1 \sigma_2 \dots \sigma_L}}{\sqrt{N}}, \\ O_2 &= \frac{\sigma_L}{\sigma_1}, \\ O_3 &= \sigma_L, \\ O_4 &= \frac{\sigma_1^2}{\sigma_L}, \\ O_5 &= \frac{1}{\frac{1}{\sigma_1} + \frac{1}{\sigma_2} + \dots + \frac{1}{\sigma_L}}, \end{aligned} \quad (15)$$

where  $\sigma_1, \dots, \sigma_L$  are the singular values in descending order obtained by singular value decomposition of the Jacobians. The indices are scaled to  $O_i \in [0, 1]$ .

The robot cell, including the calibration object, is implemented in the simulation framework Gazebo as depicted in Fig. 2. We use the Robot Operating System (ROS) and the motion planning framework MoveIt [33] to control the robot and switch seamlessly between simulation and real-world application. Our robot path planning algorithm considers constraints for the sensor's field of view and avoids collisions with itself and its environment. In the beginning, we acquire a dataset of approximately 2,500 measurements with a randomly chosen robot configuration satisfying the previously introduced constraint for both translation and rotation. We use the dataset to train the algorithm in the subsequent section.

### B. The method

The simulated measurements of a planar and a spherical object build a pool of potential sensor poses and corresponding measurements  $\mathcal{P} = \{p_1, \dots, p_N\}$ . The subsequently described algorithm works for calibration of both orientation and translation. Given an initial set of  $n$  poses  $\mu_0 = \{p_{i0}, \dots, p_{in}\} \in \mathcal{P}$ , the first part of the algorithm adds a pose  $p_{i+}$  to the current set  $\mu$ .

Therefore, we randomly draw a pose candidate  $p_{i*}$  from the pool  $\mathcal{P}$  and calculate the observability of the extended set  $\mu = \mu \cup \{p_{i*}\}$ . The parameter  $U$  defines the number of iterations for this sampling process. The algorithm chooses the pose  $p_{i+}$  providing the highest information growth and adds it to the current view pose set  $\mu$ . The other poses are put on a so-called Tabu list to ignore them in the next sampling run and increase sampling efficiency. The parameter  $U$  increases from an initial value of 300 by 20 if the chosen observability index of the novel set did not improve by more than 0.0001 for 3 iterations.

The Tabu list prevents the algorithm from being stuck in local optima by removing poor measurement poses from the pool. If the Tabu list reaches its maximum size, a random sample of poses is removed from the Tabu list and put back to the set of potential view poses. That prevents the algorithm from removing good poses in the beginning, which become relevant in a later stage of the optimization. This procedure is similar to the approach of [21] which adds newly generated poses to the set instead. Although this might increase the variance in poses, it raises practical problems of simulating measurements in parallel to the optimization procedure.

After selecting a new view pose, the algorithm removes a pose  $\mu_{i-} \in \mu \cup \mu_{i+}$ , such that the observability index of the remaining set is maximized among all indices  $i_- \in \{1, \dots, n\}$ . In other words, we remove the worst view pose contributing the least information. The algorithm repeats the addition and removal of poses for a predefined number of iterations  $K$ . Fig. 5 displays the whole approach.

### V. EXPERIMENTAL EVALUATION

For evaluating our hand-eye calibration approach, we take real-world laser scans from 12 sensor orientations for both components. The experimental setup consists of a Universal Robots UR10e equipped with a welding gun and a Micro-Epsilon scanCONTROL 3000-100 laser profile scanner.

The chosen view poses are structured regularly such that the selection considers all degrees of freedom while simultaneously taking the sensor field of view into account. We take between 20 and 30 scans for each viewing angle of the planar measurements. The CAD models of the sensor and its mechanical mount provide an initial guess for the transformation. To determine the performance and accuracy of our approach, we use the pairwise distances between the detected sphere centers from Section III-C described in Equation (13). Since the spheres have a fixed position, the distance of all fitted center points should be as small as possible. Fig. 6a gives a comparison between the initial guess and the transformation after calibration regarding this

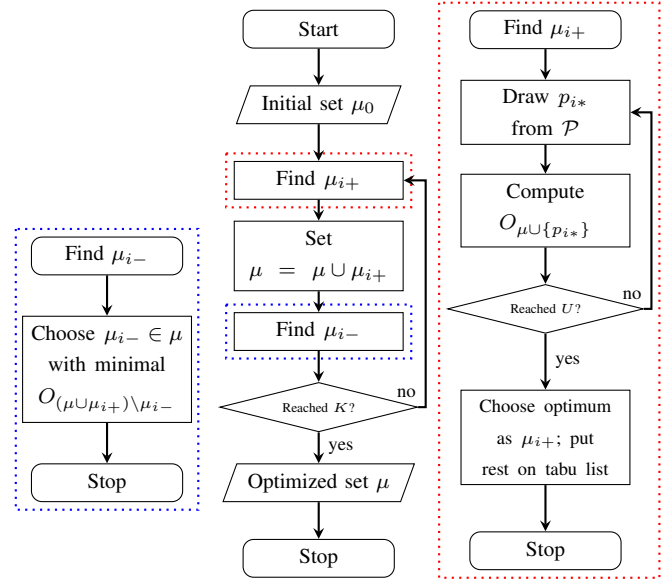


Fig. 5: The view pose optimization algorithm.

evaluation method, and Fig. 6b shows the accuracy achieved with this method.

Table II shows the performance of different hand-eye calibration approaches. Our approach achieved a median accuracy of 0.3 mm ( $x$ ,  $y$ , and  $z$  dimensions combined) while the work of [4] achieves a standard deviation of reconstructed data points of 0.05 mm. The latter uses a more complex method resulting in better overall results. [10] achieve a mean tool position error of 0.19 mm, but require a specific tool as well as a prepared calibration plane. The approach of [13] proposes the error of the transformation from sensor TCP to flange as an metric and achieve a median accuracy of 0.3 mm and  $0.1^\circ$ . However, this metric is not measurable and only available in simulation. Although the authors tested their method on a real robot, they did not provide a precise value for accuracy.

Comparing these results is only possible to a limited extent, because they were achieved on different setups regarding hardware as well as environmental impacts. Additionally, since there is no common metric of accuracy, most approaches define their own metrics which are often similar but not identical.

For testing the view pose optimization described in Section IV, we take the dataset generated in simulation as the training set. The previously described real-world measurements act as a validation set to compare the different methods and view pose selection strategies. Fig. 7 displays the experimental results for view pose selection based on our approach. The experiments in simulation revealed an improvement from approximately 0.05 mm up to approximately 0.01 mm in terms of the previously described metric. While the accuracy seems promising in simulation, the real world performance still needs to be analyzed because comparison between simulation and real world is not feasible.

TABLE II: Comparison of hand-eye calibration algorithms for 2D laser profile scanners.

| Method                    | Metric               | Accuracy      | Setup                                   |
|---------------------------|----------------------|---------------|-----------------------------------------|
| Sharifzadeh et al. [4]    | Standard deviation   | 0.05 mm       | Fanuc LR Mate 200 iC + scanCONTROL 2900 |
| Lembono et al. [10]       | Mean TCP error       | 0.19 mm       | DENSO VS060 + scanCONTROL 2600          |
| Carlson et al. [13]       | Transformation error | 0.3 mm & 0.1° | ABB IRB140 + Meta SLS 25                |
| Present w/o optimization  | Median distance      | 0.3 mm        | UR10e + scanCONTROL 3000                |
| Present with optimization | Median distance      | 0.01 mm       | UR10e + scanCONTROL 3000 (Simulation)   |

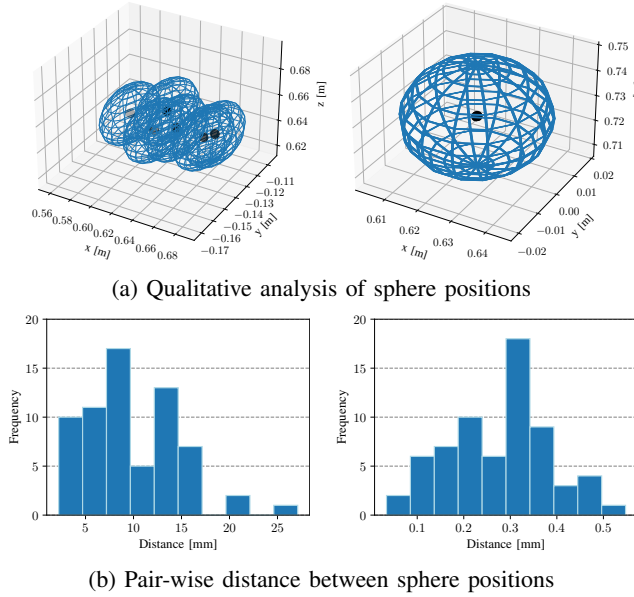


Fig. 6: Accuracy before and after calibration of a real robot cell. Sphere centers are transformed into the world frame and then used to estimate the accuracy of the method.

## VI. DISCUSSION

In general, Section V shows promising results though it is hard to compare because other approaches use varying experimental setups and suffer from different environmental influences.

Section I outlined four main goals to be fulfilled by the calibration procedure: Our calibration artifacts accomplish simplicity in terms of usability and interaction, as both objects are common household items. Additionally, the calibration method does not require complicated fitting algorithms and can be implemented using common libraries in point cloud fitting without expert knowledge.

Following [4], the separated two-step approach leads to error propagation from the orientational to the translational component. However, the algorithms revealed sufficiently good results that might come from the fact that planar fits are especially suited for calculating the orientations and spherical fits provide good results for the translational part. Nevertheless, calibrating both components using either a single plane or a sphere is part of our ongoing research.

In addition to the performance in terms of accuracy,

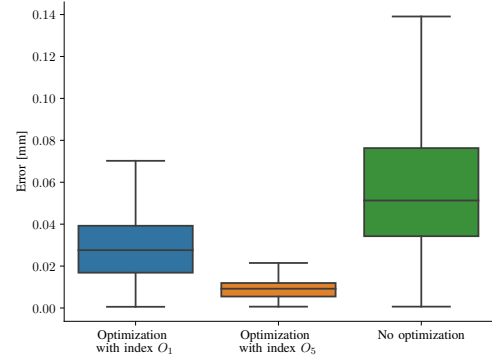


Fig. 7: Accuracy of the hand-eye calibration algorithm for regularly chosen poses and our optimization strategy with indices  $O_1$  and  $O_5$ .

the presented approach can be highly automated regarding programming, measuring, and computation. This is highly beneficial from a practical point of view. In our setup, manually generating the calibration program usually took between one and several hours, depending on the programming skills of the user, while the automatic calibration requires about 10 minutes for setup and 10 minutes for the calibration process.

A critical aspect for view pose selection may be the strict separation between data acquisition, optimization, and transfer to the robot cell. Our future work will target this problem by generating measurements for new poses on the run and enhancing the sim-to-real transfer [21].

## VII. CONCLUSION

The authors presented a novel method for hand-eye calibration of robot-guided 2D laser profile scanners that accomplishes simplicity in terms of calibration artifacts, data acquisition, and easy-to-implement algorithms.

The presented approach is already being used to calibrate a customer's welding robot. In particular, this customer benefits from not requiring expert knowledge as well as the almost fully automated procedure. The accomplished accuracy is sufficient for stitching 2D scans into 3D point clouds used in 3D feature detection pipelines in robotic applications like welding. Moreover, the view pose optimization procedure leads to even more accurate results required for other applications. Future work will further speed up data acquisition, point cloud fitting, and view pose optimization.



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